

SAEED AHMED KHAN

Ph.D. · Senior Member IEEE · National Certified Reviewer (Sindh HEC)

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PROFESSIONAL PROFILE

Engineering academic and materials–sensing researcher advancing flexible, wearable, self-powered, and thermogalvanic sensing systems while leading engineering education, accreditation, and quality-enhancement at institutional scale.

A senior engineering academic with 18+ years across higher education, applied research, and engineering practice in five countries — Pakistan, China, South Korea, Croatia (EU), and trans-regional collaborations. The research program bridges nanomaterials, flexible electronics, and self-powered sensing, with ~1,000 citations across 60+ publications in venues including ACS Sensors, Nano Energy, Chemical Engineering Journal, Journal of Power Sources, Journal of Materials Chemistry C, and Advanced Engineering Materials. The leadership portfolio spans dean-level engineering faculty leadership, quality assurance and accreditation, outcome-based education, HEC and international rankings, journal editorship, and graduate supervision.

ACADEMIC & PROFESSIONAL EXPERIENCE

Dean, Faculty of Engineering and Technology

January 2026 – Present

Sukkur IBA University · Sindh, Pakistan

Provides academic, strategic, and administrative leadership for the Faculty of Engineering and Technology, encompassing electrical engineering and allied engineering programs. Sets the academic vision, oversees curricula and accreditation, develops faculty, scales research, manages resources, and represents the Faculty within the University and with national and international partners.

Key responsibilities and contributions:

- Provide overall academic, administrative, and strategic leadership of the Faculty of Engineering and Technology, including the Department of Electrical Engineering and allied engineering programs.
- Develop and implement the Faculty's strategic plan in alignment with the University vision, HEC Pakistan policies, and international engineering education benchmarks.
- Lead the design, review, approval, and continuous improvement of undergraduate and postgraduate engineering curricula in line with outcome-based education (OBE) principles and accreditation frameworks (PEC, Washington Accord, NCEAC).
- Oversee Faculty-wide program-level accreditation, self-assessment reporting, and accreditation evidence portfolios.

- Direct faculty hiring, mentoring, performance evaluation, tenure progression, and professional development across the engineering programs.
- Set the research agenda for the Faculty: research-grant strategy, lab development, equipment investment, industry-funded research, and inter-disciplinary collaboration.
- Foster a culture of high-quality publication output in well-indexed journals and conferences, and align Faculty research with the institutional ranking strategy.
- Build and maintain national and international academic partnerships, exchange programs, joint research initiatives, dual-degree explorations, and visiting-scholar arrangements.
- Chair and serve on statutory bodies and committees including Faculty Council, Board of Faculty, Academic Council, BASR (Board of Advanced Study and Research), Syndicate, and Senate.
- Oversee the Faculty's annual academic calendar, examinations, results, thesis defenses, and academic-integrity protocols.
- Manage Faculty operations: budgeting, resource allocation, infrastructure planning, laboratory safety, equipment procurement, and laboratory-management policies.
- Lead the integration of digital teaching tools, the Learning Management System (LMS), Course Management System (CMS), and the Enterprise Resource Planning (ERP) system across Faculty operations.
- Champion gender inclusion, diversity, accessibility, and student wellbeing initiatives within the Faculty.
- Develop industry advisory boards, employability pathways, internships, capstone-industry partnerships, and engineering practicum strategy.
- Steward Faculty-level continuous improvement loops driven by graduate-attribute attainment, OBE assessment data, employer surveys, and alumni feedback.
- Represent the Faculty externally with HEC Pakistan, Sindh HEC, Pakistan Engineering Council (PEC), accreditation councils, sister universities, industry partners, and international engineering education networks.
- Provide editorial and advisory oversight of Faculty-affiliated journals, conferences, and engineering competitions.

Director, Quality Enhancement Cell (QEC)

June 2023 – January 2026

Sukkur IBA University · Sindh, Pakistan

Held the distinction of being a National Certified Reviewer of Higher Education Institutions, a credential conferred after successfully completing a rigorous 100-hour formal review training program of Sindh HEC. Directed the institutional quality-assurance system and led all aspects of program self-assessment, accreditation evidence, OBE training, continuous improvement, and ranking submissions.

Key responsibilities and contributions:

- Oversaw and enhanced the quality of academic programs, administrative services, and overall institutional performance across the University.
- Developed and implemented strategic plans for institutional quality enhancement, quality assurance, and continuous improvement.
- Established and maintained quality benchmarks, Key Performance Indicators (KPIs), and program-level performance metrics aligned with HEC Pakistan and international standards.

- Designed and implemented quality assurance frameworks, manuals, processes, and review templates across academic and administrative units.
- Conducted regular internal quality audits, program reviews, and assessment cycles in collaboration with departments, faculties, and units.
- Oversaw the preparation, validation, and submission of program-level Self-Assessment Reports (SARs) to HEC Pakistan.
- Developed and refined University policies and Standard Operating Procedures for quality assurance, OBE, examinations, and academic integrity.
- Monitored and evaluated the implementation of approved policies across faculties and recommended evidence-based improvements.
- Collected, analyzed, and reported data on academic performance, student feedback, course-evaluation outcomes, attainment of program learning outcomes, and institutional effectiveness.
- Identified areas for improvement using data-driven insights and led targeted corrective-action plans.
- Ensured continuous compliance with HEC Pakistan accreditation requirements, NCEAC engineering accreditation standards, and international quality assurance practices.
- Coordinated accreditation and re-accreditation processes for academic programs and the institution.
- Organized faculty and staff training programs on OBE, quality assurance, SAR preparation, rubric design, and reviewer practice.
- Promoted a culture of continuous improvement and quality enhancement across academic and administrative communities.
- Liaised with internal and external stakeholders including faculty leadership, HEC Pakistan, Sindh HEC, NCEAC, PEC, accreditation visiting teams, and partner universities.
- Facilitated communication and collaboration between departments to embed quality processes within academic operations.
- Prepared and presented periodic quality assurance reports to the Vice Chancellor, Senate, Syndicate, BASR, Academic Council, and statutory bodies.
- Maintained comprehensive documentation of quality assurance processes, policies, manuals, evidence, and outcomes.
- Identified and incorporated national and international best practices in quality assurance and quality enhancement.
- Fostered a culture of innovation, evidence-based decision-making, and continuous improvement within the academic ecosystem.
- Provided leadership and direction to the QEC team, distributing responsibilities and mentoring staff on review practice and data analytics.
- Managed the QEC's annual budget, training budget, and resources efficiently.
- Ensured full compliance with the Higher Education Commission (HEC) Pakistan's guidelines for QEC operations and reviewer practice.
- Led the institution's submissions to HEC Pakistan national rankings of Higher Education Institutions (HEIs), preparing evidence dossiers, data validation, and ranking strategy.

- Led international rankings engagements including Times Higher Education (THE) Impact Rankings, THE World University Rankings, and the UI GreenMetric World University Rankings — covering SDG mapping, data audit, evidence curation, and submission management.

Secretary, SIBA Testing Services (STS)

November 2024 – February 2025

Sukkur IBA University · Sindh, Pakistan

Served as Secretary of SIBA Testing Services, the University's national-scale testing wing responsible for conducting standardized admission and recruitment tests across public-sector and academic clients in Pakistan.

Key responsibilities and contributions:

- Provided executive oversight of STS testing operations including test design, item-bank management, candidate registration, and large-scale logistics.
- Coordinated with client institutions (universities, public-sector organizations, and recruitment bodies) on test specifications, scheduling, and reporting.
- Ensured strict adherence to test-security protocols, examination ethics, and confidentiality at every stage of test development and delivery.
- Supervised the secretariat's administrative, financial, and human-resource functions to deliver tests within budget and on schedule.
- Maintained compliance with applicable regulatory, institutional, and contractual requirements for testing services.
- Drafted and reviewed governance documents, MoUs, service-level agreements, and operational SOPs for STS.
- Prepared statistical and operational reports for University leadership on test cycles, candidate volumes, and client outcomes.

Associate Professor

August 2019 – January 2026

Sukkur IBA University, Department of Electrical Engineering · Sindh, Pakistan

Held the rank of Associate Professor in the Department of Electrical Engineering. Combined research leadership in flexible sensors and self-powered devices with statutory governance, editorial service, and the supervision of undergraduate and postgraduate research.

Key responsibilities and contributions:

- Designed and delivered core and elective courses across electronics, devices, sensors, control, instrumentation, and nanotechnology in the undergraduate and postgraduate engineering programs.
- Served as Member of the University Syndicate against the position of Associate Professor for a three-year term commencing October 2021.
- Led the research line on flexible nanomaterial-based sensors, thermogalvanic hydrogels, and triboelectric nanogenerators, producing publications in high-quality international journals.
- Supervised undergraduate Final Year Projects and Master of Engineering theses with publications in indexed venues.
- Worked as Chief Editor, Sukkur IBA Journal of Emerging Technologies (SJET), an HEC-recognized journal in Y category, from 2018 to 2022 — driving editorial policy, reviewer recruitment, and publication ethics.

- Served as Member of the Departmental Strategic Committee, reviewing academic, research, and co-curricular activities.
- Served as Member of the Departmental Outcome-Based Education (OBE) Accreditation Committee for the review of undergraduate engineering curricula.
- Served as Member of the Faculty/Departmental Council for the review of undergraduate and postgraduate program curricula.
- Served as Member of the Program Review Committee of the Quality Enhancement Cell for the assessment of M.E./M.S./Ph.D. self-assessment reports.
- Served as Member of the University Plagiarism Committee, Board of Studies, Board of Faculty, Academic Council, Board of Advanced Study & Research (BASR), Syndicate, and Senate.
- Served as Member, Board of Studies, Department of Electronics, University of Sindh, Jamshoro.
- Served as Member, Board of Faculty Council, Benazir Bhutto University of Technology and Skill Development.
- Served as Subject Expert (Electronics) for evaluating candidates for vacant academic and research positions at Sukkur IBA University.
- Organized the 1st, 2nd, and 3rd International Conference on Computing, Mathematics and Engineering Technologies (iCoMET) as General Chair and Co-General Chair.
- Mentored junior faculty on research-grant writing, publication strategy, and OBE-aligned course design.

Research Professor / Postdoctoral Fellow

May 2022 – April 2023

Sungkyunkwan University (SKKU), School of Mechanical Engineering · Suwon, South Korea

Conducted postdoctoral and research-professorial work at SKKU, advancing electrohydrodynamic (EHD) jet-printing of soft electronics, PDMS-based flexible devices, and triboelectric nanogenerator architectures.

Key responsibilities and contributions:

- Designed, fabricated, and characterized flexible electronic devices using high-resolution electrohydrodynamic (EHD) jet printing of PDMS and conductive inks.
- Developed multi-mode triboelectric nanogenerators (contact-mode and slide-mode TENGs) and hybrid energy-harvesting systems integrating photovoltaic and triboelectric modules.
- Investigated electrohydrodynamic underfill flow, capillary dynamics between parallel plates, and the impact of solder-bump geometry on flow under EHD effects.
- Co-authored international peer-reviewed publications in *Polymers*, *Energy Reports*, *Heliyon*, *Sensors and Actuators A: Physical*, and *Physica Scripta*.
- Strengthened cross-border research links between South Korean, Chinese, and Pakistani research groups, supporting follow-on joint publications and exchange.
- Mentored graduate students on device fabrication protocols, EHD jet printing parameters, materials characterization, and data analysis.

Assistant Professor**July 2012 – August 2019***Sukkur IBA University, Department of Electrical Engineering · Sindh, Pakistan*

Designed and delivered core and specialized engineering courses, established the research foundation in flexible sensors and nanomaterials, and built the institutional research infrastructure during a period of significant program growth.

Key responsibilities and contributions:

- Designed and delivered specialized core courses across electronics, instrumentation, control, communications, and nanotechnology in the undergraduate engineering program.
- Established the experimental research line on flexible sensors based on carbon nanomaterials, including device fabrication, materials synthesis, and electrical characterization.
- Secured and led the HEC Pakistan Start-up Research Grant (2018) as Principal Investigator on Flexible Sensors based on Nanomaterials.
- Served as Co-Principal Investigator on the HEC Pakistan Start-up Research Grant (2019) on supporting paralysis patients through EEG signal processing (Brain-Computer Interface).
- Mentored undergraduate Final Year Projects in flexible electronics, energy harvesting, biomedical signal processing, and embedded electronics.
- Organized and chaired the 1st, 2nd, and 3rd International Conference on Computing, Mathematics and Engineering Technologies (iCoMET) as General Chair and Co-General Chair.
- Drove laboratory development, equipment procurement, and graduate-attribute-aligned lab manuals for foundational and advanced electronics laboratories.
- Coordinated the Department's accreditation-readiness activities, course portfolios, OBE rubric design, and PLO–CLO attainment processes.

Lecturer**January 2012 – July 2012***ISRA University · Hyderabad, Pakistan*

Delivered undergraduate electronics and electrical engineering courses, supported laboratory instruction, and contributed to course-portfolio preparation.

Key responsibilities and contributions:

- Delivered undergraduate lectures and laboratory sessions in electronics, circuit analysis, and electronic devices.
- Prepared lecture notes, assessment instruments, lab manuals, and assignment problem sets aligned with the program curriculum.
- Mentored undergraduate students on course projects and basic research investigations.
- Supported program-level activities including examinations, assessment, and student counseling.

Research Assistant**September 2010 – July 2011***University of Zagreb, Faculty of Electrical Engineering and Computing · Zagreb, Croatia (European Union)*

Erasmus Mundus research stay focused on advanced silicon bipolar transistor characterization, supporting the M.E. thesis on the electrical properties of fabricated transistor regions in advanced silicon bipolar technologies.

Key responsibilities and contributions:

- Conducted electrical characterization of fabricated transistor regions in advanced silicon bipolar technologies using semiconductor parameter analyzers.
- Analyzed device-level current-voltage, capacitance-voltage, and small-signal characteristics to extract device parameters.
- Prepared technical reports, presentations, and the thesis manuscript under the supervision of host-laboratory faculty.
- Built international research literacy and collaboration skills across EU and Asian engineering research environments.

Instrumentation Engineer

February 2008 – May 2009

Habib Sugar Mills Ltd. · Sindh, Pakistan

Industrial instrumentation and control engineering role in a process plant environment, responsible for the reliability of measurement, control, and automation systems.

Key responsibilities and contributions:

- Maintained and calibrated process instrumentation including pressure, temperature, level, and flow measurement loops.
- Tuned and supported PID control loops, DCS/PLC systems, motor-protection equipment, and field-instrumentation networks.
- Implemented preventive-maintenance schedules, instrumentation troubleshooting, and root-cause analysis of process-control issues.
- Coordinated with operations, electrical, and mechanical teams during shutdowns, start-ups, and capacity-improvement initiatives.
- Documented instrument data sheets, loop drawings, and maintenance records in compliance with plant procedures.

Assistant Works Manager

April 2007 – January 2008

Government of Pakistan · Pakistan

Public-sector engineering management role with responsibility for technical operations, supervision of staff, and execution of engineering works under government standards and procedures.

Key responsibilities and contributions:

- Supervised engineering operations, project execution, and maintenance activities under government works procedures.
- Managed teams of technicians and skilled labor, ensuring safety, quality, and schedule compliance.
- Prepared technical estimates, work orders, and progress reports in alignment with public-sector procurement and audit requirements.
- Coordinated with vendors, contractors, and inter-departmental teams to deliver engineering deliverables.

Electrical Engineer

January 2006 – December 2006

Khairpur Sugar Mills Ltd. · Khairpur, Sindh, Pakistan

Entry-level electrical engineering role in a heavy-process industry, responsible for the safe operation and maintenance of electrical systems supporting plant operations.

Key responsibilities and contributions:

- Operated and maintained low- and medium-voltage power distribution systems, including switchgear, transformers, and motor-control centers.
- Performed routine and preventive maintenance, electrical protection coordination, and fault analysis for plant equipment.
- Supported plant electrical safety, lock-out / tag-out compliance, and grounding-system audits.
- Worked alongside instrumentation and mechanical teams to deliver reliable plant operations through season cycles.

EDUCATION

Doctor of Philosophy (Ph.D.)

2017

University of Electronic Science and Technology of China (UESTC) · Chengdu, P.R. China

Thesis: The Design and Construction of Flexible Sensors Based on Carbon Nanomaterials

- Awarded under the Chinese Government Scholarship.

Master of Engineering — Electronic Systems Engineering

2011

Mehran University of Engineering and Technology, Jamshoro · Pakistan · Research stay: University of Zagreb, Croatia (EU)

Thesis: Electrical Properties of Fabricated Transistor Regions in Advanced Silicon Bipolar Technologies

- Erasmus Mundus Scholarship recipient — University of Zagreb.

Bachelor of Engineering — Electronics Engineering

2005

Mehran University of Engineering and Technology, Jamshoro · Pakistan

Thesis: Image Edge Detection and Object Recognition Using Wavelet Transform

RESEARCH INTERESTS

- Sensors and sensing technologies (flexible, wearable, mechano-optical, biomedical).
- Flexible and wearable electronic devices on polymeric and gel substrates.
- Electronic thin films, carbon nanomaterials, and printed conductive structures.
- Triboelectric and hybrid nanogenerators for self-powered systems.
- Thermogalvanic hydrogels, organohydrogels, and gel-based energy conversion.
- Energy-harvesting systems integrating contact-mode TENG, slide-mode TENG, and photovoltaic modules.
- Self-powered human-machine interfaces: respiration, posture, facial expression, smart pen, on-hand virtual interface.
- Electrohydrodynamic (EHD) jet printing of PDMS and conductive inks for soft electronics.

- Biomedical signal processing (EEG, drowsiness detection, medical image de-noising).
- Engineering education, outcome-based education, accreditation, and quality enhancement.

RESEARCH GRANTS & FUNDED PROJECTS

2024 — Sindh Research Support Programme (SRSP), Sindh Higher Education Commission.

Role: Principal Investigator · Amount: PKR 2.4 million.

Competitive provincial research grant supporting advanced sensors, flexible-electronic device research, and materials capability.

2019 — Start-up Research Grant Programme (SRGP), Higher Education Commission of Pakistan.

Role: Co-Principal Investigator.

Brain-computer interface research supporting paralysis patients through EEG signal acquisition, processing, and intent decoding.

2018 — Start-up Research Grant Programme (SRGP), Higher Education Commission of Pakistan.

Role: Principal Investigator.

Flexible Sensors Based on Nanomaterials — establishing the materials, fabrication, and characterization line for flexible sensors.

2013 — Sustainable Alternative-Energy Project, ICT R&D Fund (Pakistan).

Role: Investigator.

Sustainable energy system for street lighting, residential power, and road-side parking, designed to reduce reliance on thermal generation and minimize ongoing environmental impact.

MEMBERSHIPS

- Senior Member, Institute of Electrical and Electronics Engineers (IEEE) — since 2012.
- Member, Pakistan Engineering Council (PEC) — since 2006.
- National Certified Reviewer of Higher Education Institutions, Sindh HEC — credentialed through a 100-hour formal review training program.

ACADEMIC RESPONSIBILITIES & GOVERNANCE

- Member, University Syndicate, Sukkur IBA University — three-year statutory term commencing October 2021 (against the position of Associate Professor).
- Member of statutory bodies of Sukkur IBA University: Plagiarism Committee, Board of Studies, Board of Faculty, Academic Council, Board of Advanced Study and Research (BASR), Syndicate, and Senate.
- Member, Outcome-Based Education (OBE) Accreditation Committee, Department of Electrical Engineering, Sukkur IBA University — curriculum review of undergraduate engineering programs.
- Member, Departmental and Faculty Council, Faculty of Electrical Engineering, Sukkur IBA University — review of undergraduate and postgraduate program curricula.

- Member, Departmental Strategic Committee, Department of Electrical Engineering — review of academic, research, and co-curricular activities.
- Member, Program Review Committee, Quality Enhancement Cell (QEC), Sukkur IBA University — assessment of M.E./M.S./Ph.D. self-assessment reports.
- Member, Board of Studies, Department of Electronics, University of Sindh, Jamshoro.
- Member, Board of Faculty Council, Benazir Bhutto University of Technology and Skill Development — review of undergraduate curricula.
- Subject Expert (Electronics) for evaluating candidates on vacant academic and research positions at Sukkur IBA University.
- Chief Editor, Sukkur IBA Journal of Emerging Technologies (SJET) — HEC-recognized Y-category journal, 2018–2022.
- General Chair / Co-General Chair, International Conference on Computing, Mathematics and Engineering Technologies (iCoMET) — 1st, 2nd, and 3rd editions.
- Supervised six (06) undergraduate research theses.
- Supervised three (03) Master of Engineering theses, several with publications in indexed venues.

TEACHING & RESEARCH SUPERVISION

Teaching Philosophy

My pedagogy rests on three commitments. First, fundamentals you can defend — every theorem traces back to a measurement. Second, devices you can hold — every course closes with a built artifact, a measurement bench, or a working circuit. Third, research literacy from day one — students leave my classrooms able to read a high-impact paper, reproduce a measurement, and write a defensible technical report. Teaching is closed-loop: I gather attainment evidence (PLOs and CLOs), inspect it term-over-term, and adjust delivery, assessments, and laboratory experiences accordingly. This is the same continuous-improvement loop I have institutionalized as Director QEC and now extend across the Faculty as Dean.

Academic Operations Platforms

At Sukkur IBA University, three core academic-operations platforms are used end-to-end by faculty and staff: (1) Learning Management System (LMS) for managing course-related activities, lecture materials, assignments, quizzes, marks, and a structured gradebook for tracking academic progress; (2) Course Management System (CMS) for academic records, attendance, course withdrawals, and other student-related administrative processes; and (3) Enterprise Resource Planning (ERP) system for staff and HR services, payroll, leave management, and performance tracking. As Dean and former Director QEC, I have used these systems both as a user and as a leader driving training, adoption, data quality, and continuous-improvement decisions.

Courses Designed and Delivered

Sensors, Nanotechnology, and Materials: Sensors Technology; Introduction to Nanotechnology; Nanotechnology; New Energy Materials and Devices; Electronic Packaging; Optoelectronics; RF MEMS.

Electronics and Circuits: VLSI Technology; Electronic Devices and Circuits; Electronic Circuit Design (Amplifiers and Oscillators); Analog and Digital Electronics; Linear Circuit Analysis; Network Circuit Analysis.

Control, Instrumentation, and Communication: Instrumentation and Measurement; Linear Control System Engineering; Advanced Control Engineering; Analog and Digital Communication; Signals and Systems.

Computing and Human–Computer Interaction: Programming (MATLAB, Python, C++); Human–Computer Interaction (HCI).

Supervision

- Supervised six (06) undergraduate research theses in B.E. Electronic / Electrical Engineering.
- Supervised three (03) Master of Engineering theses, with publications in indexed venues.

EDITORIAL SERVICE & PEER REVIEW

Chief Editor — Sukkur IBA Journal of Emerging Technologies (SJET), HEC-recognized Y category, 2018–2022.

General Chair / Co-General Chair — International Conference on Computing, Mathematics and Engineering Technologies (iCoMET), 1st, 2nd, and 3rd editions.

Fellowship Reviewer — Paris Region Fellowship Programme (ParisRegionFP), France.

Reviewer for journals (selected)

Advanced Materials; Advanced Functional Materials; Advanced Materials Interfaces; Nano-Micro Letters; Small; NANO; ACS Sensors; Sensors; Journal of Materials Chemistry A; Journal of Materials Chemistry C; Journal of Applied Energy; Energy and Environmental Materials; International Journal of Energy Research; International Journal of Imaging Systems and Technology; IEEE Access; IEEE Transactions on Intelligent Transportation Systems; IEEE/ASME Transactions on Mechatronics; ECS Journal of Solid State Science and Technology; Materials; Materials Performance and Characterization; Applied System Innovation; Electronics; Sir Syed University Research Journal of Engineering and Technology; Journal of Materials Performance and Characterization.

Reviewer for conferences and fellowships

- International Conference on Computing, Mathematics, and Engineering Technologies (iCoMET).
- International Multi-Topic ICT Conference (IMTIC).
- Paris Region Fellowship Programme (ParisRegionFP) — international postdoctoral fellowship evaluation, France.

HONORS & AWARDS

2024 Sindh HEC Sindh Research Support Programme Grant — PKR 2.4 million as Principal Investigator.

2019 HEC Pakistan Start-up Research Grant — Co-Principal Investigator on EEG-based BCI support for paralysis patients.

2018 HEC Pakistan Start-up Research Grant — Principal Investigator on Flexible Sensors Based on Nanomaterials.

2014 Chinese Government Scholarship — full Ph.D. fellowship at UESTC, P.R. China.

2010 Erasmus Mundus Scholarship (EU) — M.E. research at the University of Zagreb, Croatia.

SELECTED PUBLICATIONS

Curated selection from 60+ peer-reviewed publications. The complete filterable archive is available on the personal website. Full author list and DOIs preserved.

Recent High-Impact (2023 – 2025)

1. Y. Li, Y. Zhang, B. Chang, S.A. Khan, X. Cui, J. Zhang, H. Zhang. "Piezoionic Effect in Gels: Multifunctional Applications from Energy Harvesting to Intelligent Sensing." *Nanoscale*.
<https://doi.org/10.1039/D5NR04389A>
2. C. Bai, Y. Yao, S.A. Khan, J. Li, F. Yi, J. Zhang, H. Shi, H. Zhang. "Configurational-Entropy-Change-Driven Enhancement of Thermoelectric Performance in PVA-Based Hydrogel Thermogalvanic Cells." *Journal of Power Sources* 665, 239060. <https://doi.org/10.1016/j.jpowsour.2025.239060>
3. Y.H. Zhang, G.S. Cui, S.A. Khan, L.H. Shi, H.Z. Zhang, K. Yang, X.S. Zhao. "Self-Powered Posture Recognition Based on Thermoelectric Hydrogel." *Chinese Journal of Polymer Science*, 1–10.
<https://doi.org/10.1007/s10118-025-3440-5>
4. R. Wei, X. Cui, S.A. Khan, C. Bai, et al. "Self-Powered Versatile On-Hand Virtual Interface Enabled by Thermogalvanic Hydrogel." *Chemical Engineering Journal*, 167739, 2025.
<https://doi.org/10.1016/j.cej.2025.167739>
5. Y. Yao, S.A. Khan, X. Cui, K. Liu, S. Wen, H. Zhang. "Hydrogel-Based Smart Contact Lens for Enhanced Functional Applications." *Biomedical Instrumentation*, 100003, 2025.
<https://doi.org/10.1016/j.bmi.2025.100003>
6. X. Cui, Y. Nie, S.A. Khan, X. Bo, N. Li, X. Yang, D. Wang, R. Cheng, Z. Yuan. "Thermoelectric Gel Enabling Self-Powered Facial Perception for Expression Recognition and Health Monitoring." *ACS Sensors*, 2024.
<https://doi.org/10.1021/acssensors.4c03099>
7. Y. Zhang, H. Wang, S.A. Khan, J. Li, C. Bai, H. Zhang, R. Guo. "Deep-Learning-Assisted Thermogalvanic Hydrogel Fiber Sensor for Self-Powered In-Nostril Respiratory Monitoring." *Journal of Colloid and Interface Science*, 2024. <https://doi.org/10.1016/j.jcis.2024.09.132>
8. X. Yang, X. Ma, Y. Niu, Y. Yao, S.A. Khan, H. Zhang, X. Cui. "Thermoelectric Hydrogels for Self-Powered Wearable Biosensing." *Nano Trends*, 100050, 2024. <https://doi.org/10.1016/j.nwnano.2024.100050>
9. S.A. Khan, S. Ali, J. Moon, A. Ali, R. ul Hassan, D.H. Cho, D. Byun. "Multi-Energy Harvesting: Integrating Contact-Mode and Slide-Mode Triboelectric Nanogenerators and Solar Technologies for Efficient Power Generation in Small Electronics." *Energy Reports* 12, 4232–4240, 2024.
<https://doi.org/10.1016/j.egyr.2024.09.034>
10. S. Sang, C. Bai, W. Wang, S.A. Khan, Z. Wang, X. Yang, Z. Zhang, H. Zhang. "Finger-Temperature-Driven Thermogalvanic Gel-Based Smart Pen for Identity Recognition, Stroke Analysis, and Grip-Posture Assessment." *Nano Energy*, 2024. <https://doi.org/10.1016/j.nanoen.2024.109366>
11. C. Tian, S.A. Khan, Z. Zhang, X. Cui, H. Zhang. "Thermoelectric Hydrogel Electronic Skin for Passive Multimodal Physiological Perception." *ACS Sensors*, 2024. <https://doi.org/10.1021/acssensors.3c02172>
12. X. Yang, Z. Zhang, S.A. Khan, L. Sun, Z. Wang, X. Cui, Z. Huang, H. Zhang. "Thermogalvanic Organohydrogel-Based Non-Contact Self-Powered Electronics for Advancing Smart Agriculture." *Journal of Materials Chemistry C*, 2024. <https://doi.org/10.1039/D3TC04133F>

13. R. Hassan, S.M. Khalil, S.A. Khan, S. Ali, H. Hussain, D.H. Cho, D. Byun. "Experimental and Numerical Study of Solder Bumps Impact on Underfill Fluid Flow under EHD Effect." *Physica Scripta*, 2023. <https://iopscience.iop.org/article/10.1088/1402-4896/acfe47/meta>
14. H. Yang, S.A. Khan, N. Li, R. Fang, Z. Huang, H. Zhang. "Thermogalvanic Gel Patch for Self-Powered Human-Motion Recognition Enabled by Photothermal–Electric Conversion." *Chemical Engineering Journal*, 145247, 2023. <https://doi.org/10.1016/j.cej.2023.145247>

Foundational & High-Cited Work (2017 – 2023)

15. J. Li, Z. Wang, S.A. Khan, N. Li, Z. Huang, H. Zhang. "Self-Powered Information Conversion Based on Thermogalvanic Hydrogel with Interpenetrating Networks for Nursing Aphasic Patients." *Nano Energy*, Vol. 113, 108612, 2023. <https://doi.org/10.1016/j.nanoen.2023.108612>
16. M. Ali, S.A. Khan, A. Ali, S. Ali, R. ul Hassan, D.H. Cho, D. Byun. "Low-Profile Wind Savonius-Turbine Triboelectric Nanogenerator for Powering Small Electronics." *Sensors and Actuators A: Physical*, 114535, 2023. <https://doi.org/10.1016/j.sna.2023.114535>
17. N. Li, S.A. Khan, K. Yang, K. Zhuo, Y. Zhang, H. Zhang. "A Self-Powered Thermogalvanic Organohydrogel-Based Touch-to-Speech Braille Transmission Interface with Temperature Resistance and Stretchability." *Composites Science and Technology*, 110077, 2023. <https://doi.org/10.1016/j.compscitech.2023.110077>
18. R.U. Hassan, S.M. Khalil, S.A. Khan, J. Moon, D.H. Cho, D. Byun. "Electric-Field and Viscous-Fluid Polarity Effects on Capillary-Driven Flow Dynamics Between Parallel Plates." *Heliyon* 9 (6), 2023. <https://doi.org/10.1016/j.heliyon.2023.e16395>
19. R. Fang, X. Li, S.A. Khan, Z. Wang, X. Cui, H. Zhang, Z.H. Lin. "Anhydrous Thermogalvanic Gel for Simultaneous Waste-Heat Recovery and Thermal Management of Electronics." *ACS Applied Polymer Materials*, 2023. <https://doi.org/10.1021/acsapm.3c00481>
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24. X. Xie, H. Zhang, S.A. Khan, et al. "A Movable-Electrode Triboelectric Nanogenerator Fabricated Using a Pencil Lead for Self-Powered Locating-Collision." *Advanced Engineering Materials* 22 (6), 2000109, 2020. <https://onlinelibrary.wiley.com/doi/full/10.1002/adem.202000109>
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Selected Conference Papers

31. A. Gaffar, H. Zhang, S.A. Khan, et al. "Optical Fiber Based Dynamic Displacement Measurement Sensor Using Polymer Optical Fiber." 2nd iCoMET 2019, Sukkur IBA University.
32. M. Ali, S.A. Khan, A.Q. Rahimoon, et al. "Triboelectric Nano-Generator Scavenging Sliding-Motion Energy." 2nd iCoMET 2019.
33. I. Hussain, S.A. Khan, A.Q. Rahimoon, et al. "Flexible Triboelectric Nanogenerator Based on Paper, PET, and Aluminum." 2nd iCoMET 2019.
34. Shamsuddin, S.A. Khan, A.Q. Rahimoon, et al. "Biomechanical Energy Harvesting by Single-Electrode-Based Triboelectric Nanogenerator." 2nd iCoMET 2019.
35. S.A. Khan, M. Gao, Y. Zhu, Z. Yan, Y. Lin. "Highly Stretchable, Sensitive, and Flexible Strain Sensors Based on MWCNTs Fabricated by Different Methods." INEC 2016.
36. H. Magsi, A.H. Sodhro, S.A. Khan, G.H. Sodhro, S. Pirbhulal. "Evolution of 5G in Internet of Medical Things." iCoMET 2018.
37. N. Zahid, A.H. Sodhro, M.B. Janjua, F.A. Chachar, G.H. Sodhro, S.A. Khan. "HARQ with Chase-Combining for Bandwidth-Efficient Communication in MIMO Wireless Networks." iCoMET 2018.

Book Chapter

38. S.A. Khan, Shamsuddin, A.Q. Rahimoon, A. Abro. "Powering Healthcare IoT Sensors Based on Triboelectric Nanogenerator and Its Applications." IGI Global, 2019.

TECHNICAL SKILLS, TOOLS & LANGUAGES

Software & Tools

- Programming: MATLAB, LabVIEW, C++, Python.
- Finite Element Analysis: ANSYS, COMSOL.
- Computer-Aided Design: SolidWorks, AutoCAD.

- Scientific data: Origin Software, EHD.

Characterization & Synthesis Techniques

- CVD, EB-PVD, SEM, XRD.
- Semiconductor Parameter Analyzer.
- 3D Printing (inkjet, EHD).

Languages

- English — fluent academic and professional use.
- Urdu and Sindhi — native.
- Mandarin Chinese — beginner.
- Croatian — beginner.

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